The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 July 2000.

INCH-POUND

MIL-PRF-19500/255K
AMENDMENT 1
27 April 2000

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING,
TYPES 2N2221A, 2N2221AL, 2N2222A, 2N2222AL, 2N2221AUA 2N2222AUA,
2N2221AUB, AND 2N2222AUB, JAN, JANTX, JANTXV, JANTXVD, JANTXVH,
JANTXVM, JANTXVR, JANS, JANSD, JANSH, JANSM, JANSR, JANHC, AND JANKC

This amendment forms a part of MIL-PRF-19500/255K, dated 10 August 1998, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, delete and substitute:

"1.1. Scope. This specification covers the performance requirements for NPN, silicon, switching transistors. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance are provided for each unencapsulated device type. Provisions for radiation hardness assurance (RHA) to four radiation levels is provided for JANTXV and JANS product assurance levels. RHA level designators "M", "D", "R", and "H" are appended to the device prefix to identify devices, which have passed RHA requirements."

1.2, delete and substitute:

"1.2 Physical dimensions. See figure 1 (similar to T0-18), figures 3 and 4 (surface mount case outlines UA and UB), and figures 5, 6, 7 and 8 (JANHC and JANKC)."


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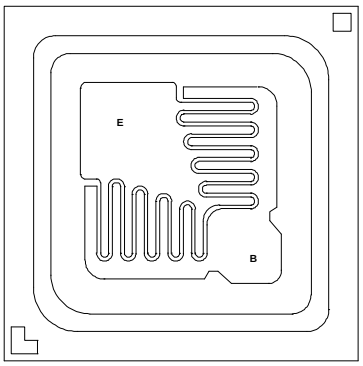
FIGURE 6, delete and substitute:

“




Physical characteristics:
B-version

| | |
|---|---|
| Chip size | 0.023 x 0.023 mils ± 0.002 mils |
| Chip thickness | 0.010 ±0.0015 mils |
| Top metal | Aluminum 15,000Å minimum, 18,000Å nominal |
| Back metal | A. Al/Ti/Ni/Ag 15kÅ/5kÅ/10kÅ/10kÅ<br>B. Gold 2,500Å minimum, 3,000Å nominal<br>C. Eutectic Die Mount - No Metal |
| Glassivation | $SI_3N_4$ 2k minimum, 2.2k nominal |
| Backside | Collector |
| Bonding pad | B = 0.0042 x 0.0042 mils, E = 0.0042 x 0.0042 mils |

FIGURE 6. JANHC and JANKC (B-version) die dimensions. ”



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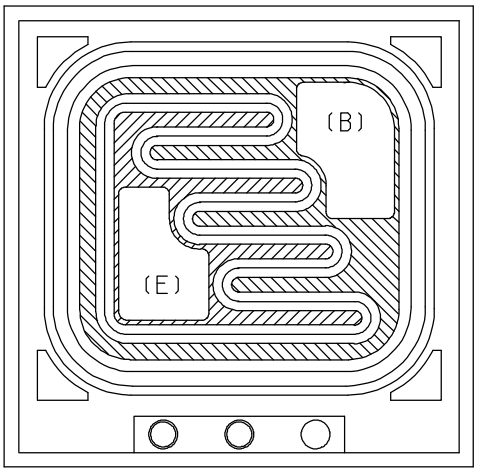
Add new figure:

"




| | |
|---|---|
| Die size. | 0.0198 x 0.0198 inches ± 0.002 inches |
| Die thickness | 0.010 ± 0.0015 inches nominal |
| Top metal | Aluminum, 21,000Å nominal |
| Back metal | A. Gold 2000Å nominal |
| Backside | Collector |
| Bonding pad | B = 0.0038 x 0.0056 inches, E = 0.0038 x 0.0056 inches |
| Glassivation | Silicon Oxide |

FIGURE 8. JANHC and JANKC (D-version for 2N2222A, and 2N2221A) die dimensions. "



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PAGE 10

3.3, second line, after “7” add "and 8".

PAGE 12

4.4.2.1, subgroup B4, conditions column, delete “$T_J$ = 150°C”.

PAGE 21

6.4, delete and substitute:

“6.4. Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2221A) will be identified on the QML.

| Die ordering information 1/ | | | | |
|---|---|---|---|---|
| PIN | Manufacturer | | | |
| | 07256 | 43611 | No source | 12969 |
| 2N2221A<br>2N2222A | JANHCA2N2221A<br>JANHCA2N2222A | JANHCB2N2221A<br>JANHCB2N2222A | N/A<br>N/A | JANHCD2N2221A<br>JANHCD2N2222A |

1/ For JANKC level, replace JANHC with JANKC. ”

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

(Project 5961-2201)



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